

VII. APPENDIX

A. Proof of Lemma 1

Assume first that $\text{rank}(A) = m$. Since A has full row rank it admits a right inverse $A^+ \in \mathbb{F}_q^{\ell \times m}$ with $AA^+ = I_m$. For an arbitrary $C \in \mathbb{F}_q^{m \times m}$ the matrix $S = A^+C$ satisfies $AS = C$, so the $\mathbb{F}_q$ -linear map $\varphi : \mathbb{F}_q^{\ell \times m} \rightarrow \mathbb{F}_q^{m \times m}$, $\varphi(S) = AS$, is surjective. A surjective linear map between finite dimensional $\mathbb{F}_q$ -vector spaces has fibers of equal cardinality $|\ker \varphi|$, hence it pushes the uniform distribution on its domain forward to the uniform distribution on its codomain, and $B = AS$ is uniform on $\mathbb{F}_q^{m \times m}$. If in addition B is invertible then

$$A(SB^{-1}y) = (AS)B^{-1}y = BB^{-1}y = y,$$

so $SB^{-1}y$ solves the system. Assume now that $\text{rank}(A) < m$. For every $S \in \mathbb{F}_q^{\ell \times m}$ the column space of AS is contained in the column space of A , so $\text{rank}(AS) \leq \text{rank}(A) < m$ and the square matrix $B = AS$ is singular, which gives $\det(B) = 0$ with probability one.

B. Proof of Lemma 2

Let $Z = (z_{ij})_{1 \leq i, j \leq s}$ be a matrix of s^2 independent indeterminates over $\mathbb{Z}$ and let $P = \mathbb{Z}[z_{11}, \dots, z_{ss}]$, an integral domain with fraction field K . The entries of $\Phi_b(Z)$ lie in P , because determinants and adjugates are polynomial maps of the entries, and so do both sides of the claimed identity. In K the element $\det(Z_{11})$ is nonzero, hence Z_{11} is invertible over K and $\text{adj}(Z_{11}) = \det(Z_{11})Z_{11}^{-1}$. Substituting this into the definition of Φ_b gives

$$\Phi_b(Z) = \det(Z_{11})(Z_{22} - Z_{21}Z_{11}^{-1}Z_{12}) = \det(Z_{11})T,$$

where T is the Schur complement of Z_{11} in Z . Taking determinants of this identity between $(s-b) \times (s-b)$ matrices over K yields $\det(\Phi_b(Z)) = \det(Z_{11})^{s-b} \det(T)$. The Schur determinant formula $\det(Z) = \det(Z_{11}) \det(T)$ holds over K as well, so $\det(T) = \det(Z) / \det(Z_{11})$ and

$$\det(\Phi_b(Z)) = \det(Z_{11})^{s-b-1} \det(Z).$$

Both sides of this equality are elements of P and they agree in K , which contains P , hence they are equal in P . Finally let R be any commutative ring and $D \in R^{s \times s}$, and let $\psi : P \rightarrow R$ be the unique ring homomorphism sending z_{ij} to D_{ij} . Determinants, adjugates and the map Φ_b are given by integer polynomials in the entries, so they commute with ψ applied entrywise, and applying ψ to the identity in P gives the statement over R .

C. Proof of Proposition 3

Apply Lemma 2 to $D^{(j)}$ with block size b_j for each $j \in \{0, \dots, L-1\}$, which gives $\det(D^{(j+1)}) = \det(D_{11}^{(j)})^{s_j-b_j-1} \det(D^{(j)})$. Multiplying these L identities and cancelling the intermediate determinants leaves

$$\det(D^{(L)}) = \det(D^{(0)}) \prod_{j=0}^{L-1} \det(D_{11}^{(j)})^{s_j-b_j-1}.$$

Since $D^{(0)} = B$ and $\delta = \det(D^{(L)})$ this is the displayed formula. The field $\mathbb{F}_q$ is an integral domain, so a product of elements of $\mathbb{F}_q$ is nonzero only if every factor is nonzero, and $\delta \neq 0$ therefore forces $\det(B) \neq 0$.

D. Proof of Lemma 4

We prove by induction on j that, conditioned on the event that $D_{11}^{(0)}, \dots, D_{11}^{(j-1)}$ are all invertible, the matrix $D^{(j)}$ is uniformly distributed on $\mathbb{F}_q^{s_j \times s_j}$. For $j=0$ this is the hypothesis on B . Suppose it holds for j and write $D = D^{(j)}$ in the block form of Lemma 2 with D_{11} of size b_j . Since D is uniform, its leading block D_{11} is uniform on $\mathbb{F}_q^{b_j \times b_j}$ and is invertible with probability α_{b_j} . Condition on an invertible value of D_{11} and on values of D_{12} and D_{21} . The map $D_{22} \mapsto D_{22} - D_{21}D_{11}^{-1}D_{12}$ is a bijection of $\mathbb{F}_q^{(s_j-b_j) \times (s_j-b_j)}$, so the Schur complement T is uniform on that space and independent of (D_{11}, D_{12}, D_{21}) . Multiplication by the nonzero scalar $\det(D_{11})$ is again a bijection of the same space, and the computation in the proof of Lemma 2 shows $\Phi_{b_j}(D) = \det(D_{11})T$ whenever D_{11} is invertible; hence $D^{(j+1)}$ is uniform on $\mathbb{F}_q^{s_{j+1} \times s_{j+1}}$, which completes the induction.

If every pivot block $D_{11}^{(j)}$ with $0 \leq j \leq L-1$ is invertible and the final matrix $D^{(L)}$ is invertible as well, then $\delta = \det(D^{(L)}) \neq 0$. Conditioning successively and using the induction above, the probability of that event is $\prod_{j=0}^{L-1} \alpha_{b_j} \cdot \alpha_{s_L}$. The quantity $\alpha_k = \prod_{i=1}^k (1 - q^{-i})$ is nonincreasing in k and $b_j \leq b$ for every j , so $\alpha_{b_j} \geq \alpha_b$ and the probability is at least $\alpha_b^L \alpha_{s_L}$. The numbers L, s_L and the sizes b_j are determined by m and b , so the bound depends only on m, q and b .

E. Proof of Theorem 5

We first establish correctness. Suppose $\text{rank}(A) = m$. By Lemma 1 the matrix $B = AS$ is uniform on $\mathbb{F}_q^{m \times m}$, so by Lemma 4 a single attempt produces $\delta \neq 0$ with probability at least $p := \alpha_b^L \alpha_{s_L}$. The κ attempts use independent multipliers S , hence they are independent, and the protocol returns $\perp$ with probability at most $(1-p)^\kappa$. If the zero test passes then $\delta \neq 0$, so Proposition 3 gives $\det(B) \neq 0$ and B is invertible; Lemma 1 then shows that $x_0 = SB^{-1}y$ satisfies $Ax_0 = y$. Define $\pi(z) = z - SB^{-1}Az$ for $z \in \mathbb{F}_q^\ell$. It is $\mathbb{F}_q$ -linear and satisfies $A\pi(z) = Az - (AS)B^{-1}Az = Az - Az = 0$, so $\text{im}(\pi) \subseteq \ker(A)$; moreover $\pi(z) = z$ for every $z \in \ker(A)$, whence $\text{im}(\pi) = \ker(A)$ and π is a linear surjection onto $\ker(A)$. A linear surjection maps the uniform distribution on its domain to the uniform distribution on its image, so $\pi(z)$ is uniform on $\ker(A)$ and $x = x_0 + \pi(z)$ is uniform on the coset $\{x : Ax = y\}$, as required by Definition 1. Suppose now that $\text{rank}(A) < m$. By Lemma 1 every attempt has $\det(B) = 0$, so Proposition 3 gives $\delta = 0$ and the zero test fails in every attempt; the protocol returns $\perp$ with probability one and the announced bit is never incorrectly equal to one.

We now describe the simulator. Let $\mathcal{C}$ be the set of corrupted parties with $|\mathcal{C}| \leq \tau$. The simulator $\mathcal{S}$ obtains from the functionality the bit β and, when $\beta = 1$, the shares of the output held by the parties in $\mathcal{C}$. Every value opened inside a multiplication is of the form $u - a$, where a is a fresh preprocessing mask drawn uniformly and independently of u and of all other masks, so its distribution is uniform on the corresponding space and independent of the remaining transcript; $\mathcal{S}$ samples all such openings uniformly at random. The number of failed attempts is sampled from the truncated geometric law with parameter p when $\beta = 1$ and set to κ when $\beta = 0$, which by the previous paragraph is the real distribution and depends only on m, q, b and

κ . For the successful attempt, $\mathcal{S}$ samples B' uniformly on $\mathbb{F}_q^{m \times m}$ conditioned on the event that the condensation chain applied to B' returns a nonzero scalar, by rejection sampling with expected number of trials $1/p$, and presents B' as the opened matrix; in a real execution B is uniform by Lemma 1 and the successful attempt conditions it on exactly that event, so the two distributions coincide. The value $\delta\rho$ is zero in each failed attempt, and in the successful attempt ρ is uniform on $\mathbb{F}_q \setminus \{0\}$ and independent of $\delta \neq 0$, so $\delta\rho$ is uniform on $\mathbb{F}_q \setminus \{0\}$; $\mathcal{S}$ samples it accordingly. Finally $\mathcal{S}$ fixes the shares of the honest parties so that they are consistent with the shares of the output supplied by the functionality, which is possible because at most τ shares are constrained and the sharing scheme is τ -private. Each message of the simulated view has the distribution of the corresponding message of the real view and the dependence structure is the same, so the two views are identically distributed and the simulation is perfect.

F. Proof of Theorem 6

We count the field elements that one party sends. Forming $B = AS$ consumes one matrix triple of shapes $m \times \ell$ and $\ell \times m$, so two masked matrices with $m\ell$ entries each are opened, which is $2m\ell$ elements. Consider the loop step j , in which the current matrix has size s_j and the pivot block has size b_j . The scalar-by-matrix product $[d][D_{22}]$ opens the masked D_{22} , of $(s_j - b_j)^2 = s_{j+1}^2$ entries, together with one masked scalar. The product $[[G]][[D_{12}]]$ opens two masked operands with b_j^2 and $b_j(s_j - b_j)$ entries, and the product $[D_{21}](G[D_{12}])$ opens two masked operands with $(s_j - b_j)b_j$ entries each. Adding these and bounding $b_j \leq b$ and $s_j - b_j \leq s_j$ gives at most $s_{j+1}^2 + 4bs_j + O(b^2)$ elements, to which the subroutine adds $\gamma_{b_j} \leq \gamma_b$. The zero test opens one element, the matrix B is opened at a cost of m^2 elements, and the two remaining products are matrix-vector products whose matrix masks are prepared in advance, so they cost $O(m + \ell)$. Summing over the loop gives the stated expression.

For the asymptotics, write $s_j = m - jb$ for $0 \leq j \leq m/b$. Then $\sum_j s_{j+1}^2 \leq \sum_{j \geq 0} (m - jb)^2 \leq m^3/(3b) + O(m^2)$ and $\sum_j 4bs_j = 4b \sum_j (m - jb) = 2m^2 + O(mb)$, while $\sum_j \gamma_b = (m/b)\gamma_b = O(mb^2)$ when $\gamma_b = O(b^3)$. Since $\ell \geq m$ the term m^2 is absorbed into $m\ell$, and the total is $O(m^3/b + mb^2 + m\ell)$.

The number of rounds is the number of loop iterations, which is $O(m/b)$, multiplied by the depth of one iteration. One iteration evaluates DetAdj on a $b \times b$ input and then performs the three products, of which the scalar-by-matrix product and the first matrix product are issued in the same round and the second matrix product follows, so its depth is that of DetAdj plus a constant. For $b \leq 4$ the subroutine has depth two, giving $O(m/b) = O(m)$ rounds, and for larger b the Samuelson–Berkowitz instantiation has depth $O(b)$, giving $O(m/b) \cdot O(b) = O(m)$ rounds as well. The steps outside the loop contribute $O(1)$ rounds.

For the local work, forming B costs $O(m^2\ell)$ field operations. Step j performs a scalar-by-matrix product on s_{j+1}^2 entries, a product of a $b \times b$ by a $b \times (s_j - b)$ matrix, a product of an $(s_j - b) \times b$ by a $b \times (s_j - b)$ matrix, and one call to DetAdj , hence $O(bs_j^2 + b^4)$ operations; summing over $O(m/b)$ steps gives $O(m^3 + mb^3)$. Opening and inverting B costs $O(m^3)$. At any moment the protocol

stores A, S , the current matrix and a constant number of masks of the same shapes, so the memory is $O(m^2 + m\ell)$ field elements.

Finally, the function $f(b) = c_1 m^3/b + c_2 m b^2$ with positive constants c_1, c_2 has derivative $-c_1 m^3/b^2 + 2c_2 m b$, which vanishes at $b = (c_1 m^2/(2c_2))^{1/3} = \Theta(m^{2/3})$, and f is convex on $b > 0$, so this is its minimum; substituting gives $f(b) = \Theta(m^{7/3})$.

G. Proof of Corollary 7

In a signing session the entry $A_{i,\cdot} = v^\top ((P_i^{(1)} + P_i^{(1)\top})O + P_i^{(2)})$ involves only the public blocks $P_i^{(1)}$ and $P_i^{(2)}$, the secret key O and the vinegar vector v , and the constant $w_i = v^\top P_i^{(1)} v$ involves the same data; none of them depends on the message. The only message-dependent quantity is the target $t = H(\mu \parallel \text{salt})$, which enters the system only through its right-hand side $y = t + w$. Every step of Π_{BDFC} reads A and the internal randomness alone, so all of them may be executed before the message is known. Writing the returned solution as

$$x = \Gamma(t + w) + \pi(z) = \Gamma t + c, \quad \Gamma = SB^{-1}, \quad c = \Gamma w + \pi(z),$$

exhibits the remaining online computation as the evaluation of the affine map $t \mapsto \Gamma t + c$ at the public point t , in which Γ and c are produced by the preprocessing.

H. Proof of Theorem 8

For the distribution of the output, the public seed is produced by a coin-flipping protocol and is therefore uniform, so expanding it yields blocks $P^{(1)}$ and $P^{(2)}$ with the distribution used by local key generation. The oil space is obtained from the random-value command of $\mathcal{F}_{\text{ABB}}$, so O is uniform on $\mathbb{F}_q^{v \times o}$ and independent of the seed. The announced block $P_i^{(3)} = \text{Up}(O^\top P_i^{(1)} O + O^\top P_i^{(2)})$ is the same deterministic function of $(P^{(1)}, P^{(2)}, O)$ that local key generation applies. Since the inputs have the same joint distribution and the function is the same, the output public key has the distribution of a locally generated Oil and Vinegar key.

For the simulation, the view of the corrupted parties consists of their shares of O and of the preprocessing material, of the values opened during the m shared products, and of the announced matrices $P_i^{(3)}$. The sharing scheme is τ -private, so the shares held by at most τ parties are distributed independently of O and can be sampled by the simulator without knowledge of O . Each opened value is of the form $u - a$ with a fresh uniform mask a , hence uniform and independent of the rest of the transcript, and is sampled uniformly. The matrices $P_i^{(3)}$ belong to the public key, which the simulator receives. It remains to fix the shares of the honest parties so that the reconstruction of each $P_i^{(3)}$ equals the value in the public key, which is possible because at most τ of the shares are already determined. The resulting view is identically distributed to the real one.

I. Proof of Lemma 9

Put $E_i = P_i^{(3)} + \text{Up}(O^\top P_i^{(1)} O + O^\top P_i^{(2)})$, an upper triangular matrix over $\mathbb{F}_q$, so that the announced key is correct exactly when $E_i = 0$ for every i . The map Up preserves the associated quadratic form, hence

$$c^\top \text{Up}(O^\top P_i^{(1)} O + O^\top P_i^{(2)}) c = (Oc)^\top P_i^{(1)} (Oc) + (Oc)^\top P_i^{(2)} c.$$

Writing $u = Oc$, $Q = \sum_i r_i P_i^{(1)}$, $R = \sum_i r_i P_i^{(2)}$ and $z = \sum_i r_i c^\top P_i^{(3)} c$, and using that the characteristic is two, we obtain

$$z + u^\top Qu + u^\top Rc = \sum_{i=1}^m r_i c^\top E_i c =: F(r, c),$$

so the verification equation $u^\top Qu + u^\top Rc = z$ holds exactly when $F(r, c) = 0$. Regard F as a polynomial in the $m+o$ indeterminates $r_1, \dots, r_m, c_1, \dots, c_o$ over $\mathbb{F}_q$; it is linear in r and quadratic in c , so its total degree is at most three. Assume $E_i \neq 0$ for some i . Expanding gives $c^\top E_i c = \sum_j (E_i)_{jj} c_j^2 + \sum_{j < k} (E_i)_{jk} c_j c_k$, and the monomials c_j^2 and $c_j c_k$ with $j < k$ are pairwise distinct, so the coefficients of this polynomial are exactly the entries of the upper triangular matrix E_i and the polynomial is nonzero. Since $r_1, \dots, r_m$ are indeterminates, the sum $F = \sum_i r_i (c^\top E_i c)$ is nonzero as well. The Schwartz–Zippel lemma bounds the probability that a nonzero polynomial of total degree at most three vanishes at a point drawn uniformly from $\mathbb{F}_q^{m+o}$ by $3/q$, which gives the soundness claim.

For the cost, the vector $u = Oc$ is a fixed public linear combination of the columns of the shared matrix O and is computed without interaction, and the same holds for Qu and for Rc , which apply public matrices to a shared and to a public vector. The only product of two shared quantities is the scalar $u^\top (Qu)$, evaluated with one Beaver triple whose two masked operands are vectors of length v , so $2v$ elements are opened, and comparing the result with the public value z opens one further element. All of these openings occur in a single round, and the count $2v + O(1)$ does not involve m or o .

J. Proof of Theorem 10

The vinegar vector is drawn by the random-value command of $\mathcal{F}_{\text{ABB}}$, so v is uniform on $\mathbb{F}_q^v$, and the matrix A and the vector w are obtained from (O, v) by the same deterministic formulas that a local signer uses. By Theorem 5 the sampler returns a vector x that is uniform on $\{x : Ax = t + w\}$ whenever it does not return $\perp$, and it returns $\perp$ only in situations in which a local signer would also draw a fresh vinegar vector. Therefore the pair (v, x) , and with it the signature $s = (v + Ox, x)$, has exactly the distribution of a signature produced by the underlying scheme.

For the online phase, Corollary 7 writes $x = \Gamma t + c$ with Γ and c produced by the preprocessing. Consequently $Ox = (O\Gamma)t + Oc$, and the two shared products $O\Gamma$ and Oc are computed during the preprocessing as well. Each party therefore obtains its share of $s = (v + (O\Gamma)t + Oc, \Gamma t + c)$ by applying a public affine map to shares it already holds, which requires no secure multiplication and no interaction. The session is completed by one broadcast of the shares of s , that is n field elements per party in one round.

For the leakage, every message of the session is either a value opened inside a multiplication, whose distribution is uniform and independent of the inputs by the argument in the proof of Theorem 5, or one of the values covered by that theorem, or the shares of the final signature. The first kind is simulatable without any input, the second is simulatable from the bit announced by the sampler, which is exactly the information that an attempt succeeded, and the third is simulatable from the signature. Hence the view of any set of at

most τ corrupted parties is simulatable from the signature and the number of failed attempts.

K. Proof of Lemma 11

Let $x_1, \dots, x_N$ be the distinct public evaluation points, let f be the polynomial of degree at most τ that defines the sharing, let $y_1, \dots, y_N$ be the received values and let $B = \{i : y_i \neq f(x_i)\}$ be the set of deviating positions, so $|B| \leq e$. For a candidate error count e' the decoder looks for a pair (Q, E) with E monic of degree e' , $\deg Q \leq \tau + e'$ and $Q(x_i) = y_i E(x_i)$ for all i , which is a system of N linear equations in the $\tau + 2e' + 1$ unknown coefficients.

Assume $2e' < N - \tau$ and let (Q, E) be any solution. Put $R = Q - fE$, a polynomial of degree at most $\tau + e'$. For every $i \notin B$ we have $Q(x_i) = y_i E(x_i) = f(x_i) E(x_i)$, so $R(x_i) = 0$, and there are at least $N - |B| \geq N - e'$ such indices. Since $N - e' > \tau + e'$, the polynomial R has more roots than its degree and is therefore zero, so $Q = fE$ and $f = Q/E$. Moreover, for $i \in B$ we get $y_i E(x_i) = Q(x_i) = f(x_i) E(x_i)$ with $y_i \neq f(x_i)$, hence $E(x_i) = 0$, so E vanishes at every deviating position and $e' \geq |B|$.

Consequently no valid pair exists for $e' < |B|$, while for $e' = |B|$ the pair $E_0 = \prod_{i \in B} (X - x_i)$ and $Q_0 = fE_0$ is a solution, because $Q_0(x_i) = f(x_i)E_0(x_i)$ equals $y_i E_0(x_i)$ both for $i \notin B$, where $y_i = f(x_i)$, and for $i \in B$, where both sides vanish. Running the decoder for $e' = 0, 1, 2, \dots$ and stopping at the first success therefore stops at $e' = |B|$, returns f and outputs $\{i : E(x_i) = 0\}$, which equals B because E is monic of degree $|B|$ and vanishes on the $|B|$ points x_i with $i \in B$. Each candidate requires solving a linear system with N equations and at most N unknowns, which costs $O(N^3)$ field operations, and at most $O(N)$ candidates are tried, giving $O(N^4)$ in total.

L. Proof of Theorem 12

Every value opened during the protocol is the difference $u - a$ of two degree- τ Shamir sharings, hence itself a degree- τ sharing, and the vector of the N shares of such a value is the evaluation of a polynomial of degree at most τ at the N distinct public points, that is a codeword of the Reed–Solomon code of length N and dimension $\tau + 1$. A coalition of at most τ corrupted signers alters at most τ coordinates of that codeword. If $\tau < N/3$ then $3\tau < N$, hence $2\tau \leq N - \tau - 1$, and Lemma 11 applies with $e = \tau$: the correct value is reconstructed and the exact set of altered coordinates is returned. This holds for every opening, so no opening fails, the honest signers complete the session and obtain the signature, which is guaranteed output delivery, and every deviating signer is named at the first opening in which it deviates. Correctness of the resulting signature follows from Theorem 10.

For the second claim, the vinegar vector and the multiplier S are both produced by the random-value command of $\mathcal{F}_{\text{ABB}}$, which by definition returns a uniformly distributed value on which no party has any influence. The matrix A is a deterministic function of the key and of v , so it is distributed as in an honest execution whatever the adversary does. By Lemma 1 the matrix $B = AS$ is then uniform whenever A has full row rank, and by Lemma 4 an attempt succeeds with probability $\alpha_b^L \alpha_{s_L}$, which is a function of m , q and b alone. Since the attempts use independent multipliers, the number of failed

attempts follows the same law as in an honest execution and the adversary cannot change it.